MATH 463 - INTEGRAL EQUATIONS

LESSON 2

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OUTLINE

1. CONVERTING INTEGRAL EQUATIONS
   - Converting Volterra Integral Equation of 1st kind to 2nd kind
   - Converting Volterra Integral Equation to Ordinary Differential Equation
   - Converting Initial Value Problem to Volterra Integral Equation

CONVERTING VOLTERRA INTEGRAL EQUATION OF 1ST KIND TO 2ND KIND

We use the Leibniz Integral Rule to differentiate the integral equation with respect to x.

$$f(x) = \int_0^x k(x,t)u(t)\ dt$$

$$\frac{d}{dx}\left\{f(x) = \int_0^x k(x,t)u(t)\ dt\right\}$$

$$f'(x) = k(x,x)u(x) + \int_0^x k'(x,t)u(t)\ dt$$

$$\frac{f'(x)}{k(x,x)} = u(x) + \int_0^x \frac{k'(x,t)}{k(x,x)}u(t)\ dt$$

$$F(x) = u(x) + \int_0^x K(x,t)u(t)\ dt$$

Example:

$$2x = \int_0^x xtu(t)\ dt$$

$$\frac{d}{dx}\left\{2x = \int_0^x xtu(t)\ dt\right\}$$

$$2 = x^2u(x) + \int_0^x tu(t)\ dt$$

$$\frac{2}{x^2} = u(x) + \int_0^x \frac{t}{x^2}u(t)\ dt$$

$$u(x) = \frac{2}{x^2} - \int_0^x \frac{t}{x^2}u(t)\ dt$$

CONVERTING VOLTERRA INTEGRAL EQUATION TO ORDINARY DIFFERENTIAL EQUATION

$$\mu(x) = f(x) + \lambda \int_{\alpha(x)}^{\beta(x)} K(x,t)\mu(t)dt$$

We will use the Leibnitz rule to convert Volterra's IE to Differential equations with initial conditions.This can be achieved by differentiating both sides of the integral equation. The differentiating process should be continued as many times as needed until we obtain a pure differential equation with the integral sign removed. Moreover the initial conditions need can be obtained by substituting x = 0 in the integral equations.

$$\frac{d}{dx}\left\{\mu(x) = f(x) + \lambda \int_{\alpha(x)}^{\beta(x)} K(x,t)\mu(t)dt\right\}$$

$$\implies \frac{d(\mu(x))}{dx} = \frac{d(f(x))}{dx} + \lambda \frac{d}{dx}\int_{\alpha(x)}^{\beta(x)} K(x,t)\mu(t)dt$$

Example

Convert the I.E below to an O.D.E

$$\frac{x^2}{2} = \int_0^x \cos(x-t)\ y(t)\ dt$$

Solution

$$\frac{d}{dx}\left\{\frac{x^2}{2} = \int_0^x \cos(x-t)\ y(t)\ dt\right\}$$

$$\implies \frac{d}{dx}\left\{\frac{x^2}{2}\right\} = \cos(x-x)\ y(x)\frac{dx}{dx} - \cos(x-0)\ y(0)\frac{d0}{dx}$$

$$+ \int_0^x y(t)\frac{d(\cos(x-t)\ )}{dx}\ dt$$

$$\implies x = y(x) - \int_0^x \sin(x-t)y(t)\ dt$$

$$\implies \frac{d}{dx}\left\{x = y(x) - \int_0^x \sin(x-t)y(t)\ dt\right\}$$

$$\implies 1 = y'(x) - \int_0^x \cos(x-t)\ y(t)\ dt$$

$$\text{but } \int_0^x \cos(x-t)\ y(t)\ dt = \frac{x^2}{2}$$

$$\implies 1 = y'(x) - \frac{x^2}{2}$$

$$y'(x) = 1 + \frac{x^2}{2}$$

$$\text{also } y(x) = x + \int_0^x \sin(x-t) y(t)\ dt$$

$$\text{when } x = 0, y(0) = 0$$

$$\therefore y'(x) = 1 + \frac{x^2}{2}, \quad y(0) = 0$$

CONVERTING INITIAL VALUE PROBLEM TO VOLTERRA INTEGRAL EQUATION

In this section we will look at the technique that converts an initial value problem to equivalent Volterra's equations. To do this we will outline a useful transformation formula.

$$\int_0^x \int_0^{x_1} \int_0^{x_2} \ldots\ldots \int_0^{x_n-1} f(x_n)dx_n \ldots dx_1 = \frac{1}{(n-1)!}\int_0^x (x-t)^{n-1} f(t)dt$$

.

that converts multiple integrals to just one single integral. For example, in the case of two and three multiple integrals we would have respectively;

$$\int_0^x \int_0^x f(t)dtdt = \int_0^x (x-t)f(t)dt$$

and

$$\int_0^x \int_0^x \int_0^x f(t)dtdtdt = \frac{1}{2!}\int_0^x (x-t)^2 f(t)dt$$

which are used to transform double and triple integrals respectively to single integrals.

Let us now apply this useful technique to the goal of this section i.e Converting Initial Value Problem to Volterra Integral Equation. Without loss of generality we apply this technique to an third order initial value problem. Consider the third order differential equation

$$y^{'''}(x) + p(x)y^{''}(x) + q(x)y^{'}(x) + r(x)y(x) = g(x)$$

subject to the initial condition

$$y(0) = \alpha, y^{'}(0) = \beta, y^{''}(0) = \gamma.$$

$$\alpha\beta\gamma$$

are constants.
Assuming that the functions $p(x), q(x)$ and $r(x)$ are analytic functions with Taylor series expansions about the origin and $g(x)$ is continuous through the interval of discussion. To transform the third order differential equation into an equivalent Volterra's Integral equation,
we first set

$$y^{'''}(x) = u(x).....(1)$$

where $u(x)$ is a continuous function in the interval of discussion. Based on (1) we find other relations for y and its derivatives as single integrals involving $u(x)$. We do this by integrating both sides of (1) from 0 to x.

$$\int_0^x y^{'''}(x)dt = \int_0^x u(t)dt$$

$$y^{''}(x) - y^{''}(0) = \int_0^x u(t)dt$$

Or equivalently,

$$y^{''}(x) = \gamma + \int_0^x u(t)dt.....(2)$$

using the initial condition $y^{''}(0) = \gamma$. To obtain $y^{'}(x)$ we integrate both sides of (2) from 0 to x to obtain

$$y^{'}(x) = \beta + \gamma x \int_0^x \int_0^x u(t)dtdt.....(3)$$

Similarly we integrate both sides of (3) from 0 to x to formally obtain

$$y(x) = \alpha + \beta x + \frac{\gamma x^2}{2} + \int_0^x \int_0^x \int_0^x u(t)dtdtdt.....(4)$$

Using the conversion formula in double and triple to reduce (3)and(4) respectively to yield single integrals gives

$$y^{'}(x) = \beta + \gamma x + \int_0^x (x-t)u(t)dt.....(5)$$

and

$$y(x) = \alpha + \beta x + \frac{\gamma x^2}{2} + \frac{1}{2}\int_0^x (x-t)^2 u(t).....(6)$$

respectively.

Now substituting $(1), (2), (5), (6)$ leads to a Volterra Integral Equation of the second kind of

$$u(x) = f(x) + \int_o^x k(x,t)u(t)dt$$

where,

$$k(x,t) = p(x) + q(x)(x-t) + \frac{r(x)}{2}(x-t)^2$$

and

$$f(x) = g(x) - [\gamma p(x) + \beta(x) + \alpha r(x) + \gamma x p(x) + r(x)(\beta x + \frac{\gamma x^2}{2})]$$

Convert the initial value problem into a Volterra integral equation;
$y''' - 3y'' - 6y' + 5y = 0, \qquad y(0) = y'(0) = y''(0) = 1$

$$Let\ y'''(x) = u(x) \qquad (1)$$

$$\int_0^x y'''(t)\ dt = \int_0^x u(t)\ dt$$

$$\implies y''(x) - y''(0) = \int_0^x u(t)\ dt$$

$$\implies y''(x) = 1 + \int_0^x u(t)\ dt \qquad (2)$$

$$y'(x) - y'(0) = x + \int_0^x \int_0^x u(t)\ dtdt$$

$$y'(x) = 1 + x + \int_0^x \int_0^x u(t)\ dtdt \qquad (3)$$

$$y(x) = 1 + x + \frac{x^2}{2} + \int_0^x \int_0^x \int_0^x u(t)\ dtdtdt \qquad (4)$$

put (1),(2),(3),(4) into the I.V.P

$$u(x) - 3(1 + \int_0^x u(t)\ dt) - 6(1 + x + \int_0^x (x-t)u(t)\ dt) + 5(1 + x + \frac{x^2}{2}$$

$$+\frac{1}{2}\int_0^x (x-t)^2 u(t)\ dt)$$

$$u(x) - 4 - x + \frac{5}{2}x^2 + \int_0^x [\frac{5}{2}(x-t)^2 - 3 - 6(x-t)]u(t)\ dt$$

$$u(x) = 4 + x - \frac{5}{2}x^2 - \int_0^x [\frac{5}{2}(x-t)^2 - 3 - 6(x-t)]u(t)\ dt$$

Exercises

1. Solve the following Volterra's IE by converting to an equivalent initial value problem and subsequently solving
   1. $$y(x) = e^x - \int_0^x (x-t)y(t)\ dt$$
   2. $$y(x) = x - \cos(x) + \int_0^x (x-t)y(t)\ dt$$
2. Convert each of the following initial value problems to their respective Volterra's integral equation
   1. $$y' + y = 0 \qquad y(0) = 1$$
   2. $$y'' + y = \sin(x) \qquad y(0) = y'(0) = 0$$